

Nekhil Nagia

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EDUCATION

University Of Toronto

Bachelor of Applied Science in Computer Engineering, Minor in Mechatronics

September 2016 – April 2021

Toronto, ON

INDUSTRY EXPERIENCE

Google

Software Engineer III

June 2023 – Present

Kitchener, ON

- Technical lead of a team of 7 working on the [Google Kubernetes Engine \(GKE\)](#) team
- Leading a project to enable a multi-tenant mode of operation of GKE leading to 10x cost savings and 5x latency improvements for SaaS customers
- Delivered a rewrite of an internal service responsible for GKE-wide administration of clusters and control plane, resulting in a monthly saving of over [200 SWE hours](#)

Amazon

Software Development Engineer II

August 2021 – May 2023

Toronto, ON

- Worked on the [AWS RDS Aurora Control Plane](#) team which is responsible maintaining and optimizing crucial control plane components powering a large fleet of databases
- [Optimized a control plane component by 2x](#) to enable down-scaling to very low resource capacity instances for Aurora Serverless v2
- Setup distributed infrastructure for billing customers based on their serverless product usage and worked on internal APIs handling [100s of requests per second](#)
- Pitched and secured funding for an efficiency project with estimated [annual savings of \\$2M](#) to re-design a key control plane agent responsible for nodes' operations
- Diagnosed and resolved critical networking issues in the service that led to customer downtime
- Actively involved in code reviews, design reviews and mentoring junior engineers in the team

Amazon

Software Development Engineer Intern

May 2020 – August 2020

Toronto, ON

- Worked on the RDS Aurora Control Plane team as an intern
- Improved Aurora QA pipeline for availability by adding over [60 failure injection and stress tests](#)
- Integrated work with software release pipelines thus preventing regressions during new releases

Square

Machine Learning Engineer Intern

February 2020 – May 2020

Toronto, ON

- Worked on the [Risk Data Science team](#) to develop solutions to mitigate fraud in Cash App
- Worked on an end-to-end deep learning pipeline to [predict chargebacks](#) using raw event level customer data, used tools such as [TensorFlow](#), [SQL](#), [Apache Airflow](#)
- Built an AI receptionist system called Deep Doorman David as a [side project](#) that uses a [custom face recognition model](#) to interact with visitors at the office and notifies the appropriate employees on Slack

Dessa

Software Engineer Intern

May 2019 – February 2020

Toronto, ON

- Built a product called Foundations for managing [end-to-end machine learning](#) lifecycle, this product is now [open-source](#)
- Involved in all areas of engineering, from setting up devops infrastructure in Jenkins & AWS to architecture and development using tools such as Kubernetes, Redis and Python
- Worked on a feature to allow the user to automatically track data health and consistency using intelligent data validation techniques and alert them to prevent model degradation in production

RESEARCH EXPERIENCE

Federated Learning for WiFi Fingerprinting

May 2021 – August 2021

- Worked under the supervision of a PhD student and Professor Valaee at [WIRLab UofT](#)
- Created a novel application of [federated learning for the Wi-Fi fingerprinting](#) problem and demonstrated [performance gains in localization accuracy by 20%](#)
- Project culminated with a research paper that was presented and [published at the IEEE ICC 2022 conference](#)

Connected Vehicles & Smart Transportation

May 2018 – May 2019

- Worked on the CVST project at the University of Toronto under a PhD student and Prof. Leon-Garcia
- Developed a high performance [web server using Golang and Python](#) to implement APIs interfacing and [processing over 100 Gigabytes](#) of data stored in Elasticsearch
- Developed a [computer vision algorithm](#) using OpenCV in C++ to process data from a real-time camera stream to [detect free parking spots](#) in a parking lot, achieved [80% accuracy](#) on test data and created interactive visualizations

PUBLICATIONS

Federated Learning for WiFi Fingerprinting

May 2022

- N. Nagia, M. T. Rahman and S. Valaee, "Federated Learning for WiFi Fingerprinting," ICC 2022 - IEEE International Conference on Communications, Seoul, Korea, Republic of, 2022, pp. 4968-4973, doi: 10.1109/ICC45855.2022.9838945.